

CURIONEST

Builder Test

Regents elements test

WHAT'S INSIDE

- ✓ Complete concept notes with worked examples
- ✓ Practice sets in varied formats (short answer, MCQ, calculations)
- ✓ Key terms glossary + full answer key with worked solutions
- ✓ Original figures and Word-native equations — print-ready & editable

Made by a chemistry teacher — aligned to the standard US chemistry sequence.

Assess Yourself:

Vocab:

Lesson 1:

Lesson 2:

Lesson 3:

Lesson 4:

LESSON 1: ATOMIC STRUCTURE**Objective:**

- Identify protons, neutrons, electrons
- Count subatomic particles in neutral atoms

The nucleus contains protons (positive) and neutrons (neutral). Electrons orbit outside the nucleus.

Worked Example

How many protons, neutrons, and electrons are in a neutral atom of ^{27}Al ($Z=13$)?

Step 1: Protons = $Z = 13$

Step 2: Neutrons = $A - Z = 27 - 13 = 14$

Step 3: Electrons = protons (neutral) = 13

Answer: 13 p^+ , 14 n^0 , 13 e^-

Check Your Understanding

1. What is the charge of a neutron?

- (A) Positive
- (B) Negative
- (C) Neutral
- (D) Variable

Answer: _____

2. How many electrons in neutral Na ($Z=11$)?

- (A) 10
- (B) 11
- (C) 12
- (D) 23

Answer: _____

Complete the Statements

- Protons are _____ charged and neutrons are _____ charged.
- The nucleus is always _____ charged.

Complete the Table (all atoms are neutral)

Atom	Protons	Electrons	Neutrons	Mass #	Atomic #
A	44			102	
B		84	125		
C				56	
D					89

Regents Practice

Answer in the style of the NY State Regents exam.

1.) The number of protons in the nucleus of ^{32}P ($Z=15$) is:

- (1) 15
- (2) 17
- (3) 32
- (4) 47

2.) What is the total number of electrons in an atom with atomic number 13 and mass number 27?

- (1) 13
- (2) 14
- (3) 27
- (4) 40

3.) Explain why atoms are electrically neutral.

Answer Key

Test